Assignment 1: Model Evaluation and Metrics

Q1. Critically analyze the limitations of classification accuracy as an evaluation metric. Using theoretical justification and examples, explain the accuracy paradox and propose alternative metrics suitable for imbalanced datasets. (10 Marks)

Q2. Derive the relationship between precision, recall, F1-score, and specificity. Discuss in detail how these metrics behave under extreme class imbalance and their implications in real-world applications such as medical diagnosis. (10 Marks)

Q3. Explain the concept of ROC curve from a probabilistic perspective. Derive how varying the decision threshold affects TPR and FPR, and discuss how ROC analysis helps in selecting an optimal classifier. (10 Marks)

Q4. Provide a detailed comparison between parametric and non-parametric models. Discuss their bias-variance trade-off, assumptions, scalability, and suitability for different types of data distributions. (10 Marks)

Q5. Explain overfitting and underfitting in the context of model evaluation. Using bias-variance decomposition, analyze how cross-validation helps in mitigating these issues and improving generalization. (10 Marks)